DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: FY B. Tech(All) **Branch: All Branches** **Semester : I**

Subject Code & Name: ENGINEERING MATHEMATICS-I **Course Code: BTBS101**

Max Marks: 60 **Date:21/07/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	If 5 is eigen value of the matrix A then eigen value of A^2 is……				Remember (CO1)	1
	a. 5	b. -5	c. 25	d. 2		
2	The rank of matrix $A=\begin{bmatrix}1&0&0\\0&1&0\\0&0&0\end{bmatrix}$				Remember (CO1)	1
	a.3	b. 2	c. 1	d. 4		
3	Homogegenous system of linear equations is				Remember (CO1)	1
	a. always consistent	b. always inconsistent	c. both a and b	d. None of these		
4	If $u = f\left(\frac{y}{x}\right)$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ is equal to				Understand (CO2)	1
	a. $2u$	b. u	c. $4u$	d. 0		
5	If $u = x^3 + y^3$ then $\frac{\partial^2 u}{\partial x \partial y}$ is equal to				Understand (CO2)	1
	a. 0	b. 3	c. $3y$	d. $3x^2$		
6	A point at which $f(x, y)$ has neither maxima nor minima is called as				Remember(CO3)	1
	a. Maxima	b. Minima	c. Saddle point	d. Extreme point		
7	The curve $x^2 - 2y = c$ is symmetrical about				Remember (CO4)	1
	a. x-axis	b. y-axis	c. y = x	d. y = -x		
8	The point of intersection with x-axis for the curve $xy^2 = 4a^2(2a - x)$				Remember (CO4)	1
	a. $(0,0)$	b. $(2a, 0)$	c. $(0,2a)$	d. $(2a, 2a)$		
9	If $u = x^2 - y^2$ and $v = 2xy$ *then* $\frac{\partial(u,v)}{\partial(x,y)}$ is equal to				Remember (CO3)	1
	a. $4(x^2 + y^2)$	b. $4x$	c. $4y$	d. None of these		

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10	The number of loops for curve $r = asin3\theta$ are				Remember (CO4)	1
	a. 2	b. 3	c. 6	d. 5		
11	The value of $\int_0^2 \int_0^y 5\, dxdy$ is.				Understand (CO5)	1
	a. 2	b. 0	c. 5	d. 10		
12	The value of $\int_0^3 \int_0^2 \int_0^1 dxdydz$.				Understand (CO5)	1
	a. 6	b. 3	c. 2	d. 1		
Q. 2	**Solve the following.**					12
A)	Find the rank of matrix $A=\begin{bmatrix} 1 & 2 & 3 & -1 \\ -1 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$ by converting it into normal form.				Understand (CO1)	6
B)	Solve the system of linear equation by testing the consistency $x+y+z=6,\ x-y+2z=5,\ 3x+y+z=8,\ 2x-2y+3z=7$				Apply(CO1)	6
Q.3	**Solve the following.**					12
A)	If $u=\sin^{-1}(x^2+y^2)$ then show that $x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+y^2\frac{\partial^2 u}{\partial y^2}=2\tan u(2\sec^2 u-1)$				Understand (CO2)	6
B)	If $u=f(y-z,z-x,x-y)$ show that $\frac{\partial u}{\partial x}+\frac{\partial u}{\partial y}+\frac{\partial u}{\partial z}=0$				Understand(CO2)	6
Q. 4	**Solve Any Two of the following.**					12
A)	Find the maxima or minima for the function $x^2+y^2+(30-x-y)^2$				Understand(CO3)	6
B)	Find the maximum value of $x^2+y^2+z^2$ when $x+y+z=3a$				Understand(CO3)	6
C)	Find the Jacobian $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ of $u=x^2-2y,\ v=x+y+z,\ w=x-2y+3$				Understand(CO3)	6
Q.5	**Solve Any Two of the following.**					12
A)	Identify and trace the curve $x=a(\theta-\sin\theta),\ y=a(1-\cos\theta)$.				Apply(CO4)	6
B)	Identify and trace the curve $r=2(1-\cos\theta)$				Apply(CO4)	6
C)	Solve $\int_0^2 x^3\sqrt{(2-x)}dx$				Apply(CO4)	6
Q. 6	**Solve Any Two of the following.**					12
A)	Solve $\iint xy^2 dxdy$ over the region bounded by $y=x^2, y=0, x=1$				Apply (CO5)	6
B)	Solve $\int_0^2 \int_0^x \int_0^{x+y} e^{x+y+z}dxdydz$				Apply(CO5)	6
C)	Find the inverse of the matrix A by Cayley-Hamilton theorem $A=\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$				Understand(CO1)	6
	***** End *****					

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